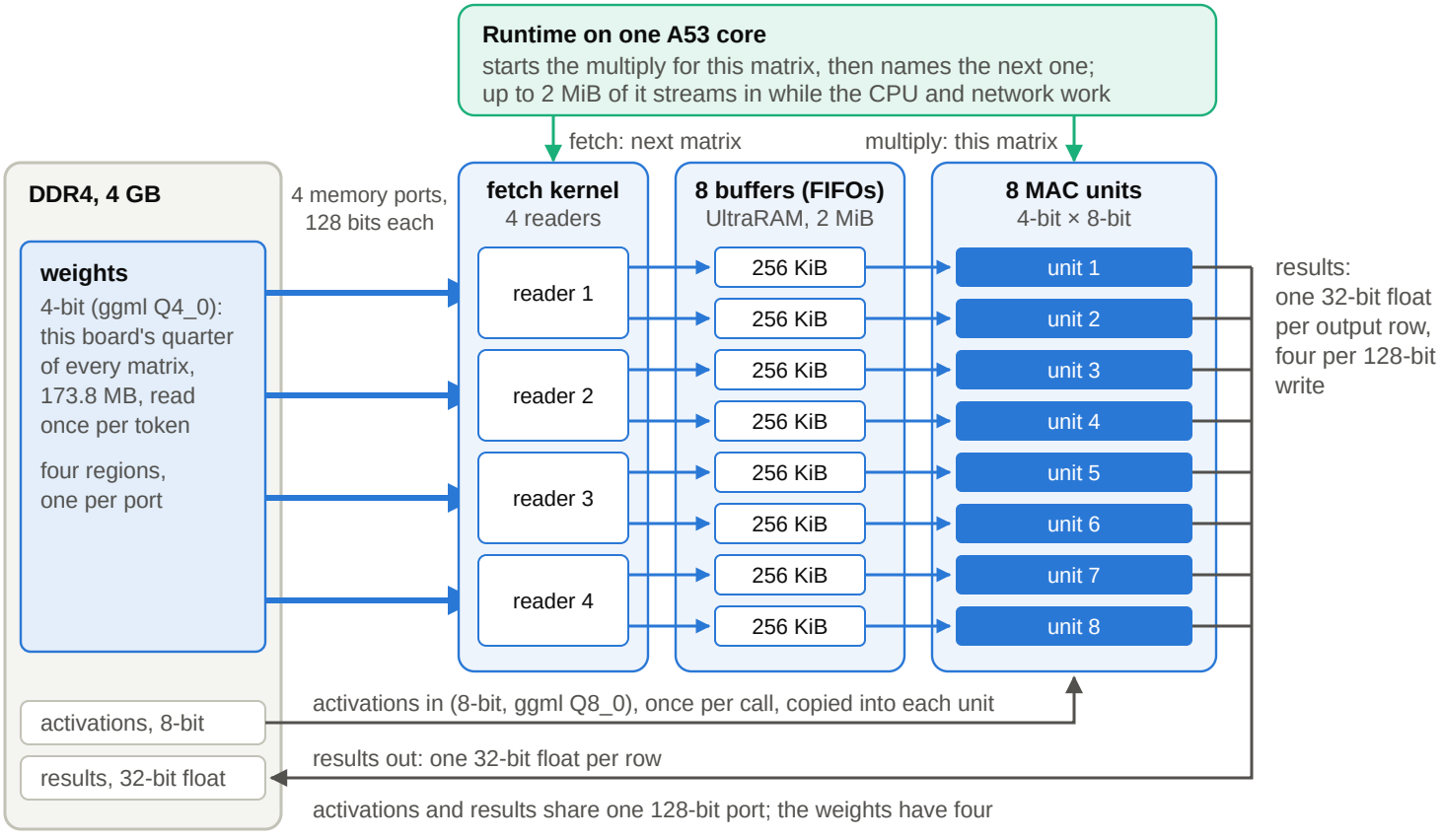
The engine on one board

Every token, the FPGA streams this board's quarter of the weights from DDR4 and multiplies it as it passes



Inside each MAC unit, for every block of 32 weights

32 × (4-bit × 8-bit) → integer sum → × two scales, 32-bit float → add into lane (block mod 8) → end of row: 8 lanes, fixed tree

The integer sum is exact in any order and the float order is fixed by the contract (src/llm_kv260/q4_0.py), so the output is checked against it byte for byte: 9 of 9 test cases pass on each of the four boards (bench_gemv2).

gemv2 on the XCK26 (Vitis HLS and Vivado 2022.2), 250 MHz clock. Each reader deals alternate 144-byte chunks to its two buffers. Sources: hls/gemv2/, hw/kv260_gemv2/build.tcl, board/tp/tp_llama.cpp. Weights are a load-time copy of the GGUF's Q4_0 blocks.